

What to take to the desktop machine

Castelnuovo di Porto nonlinear time-history · target machine: Intel i9-9900KF, 8 cores, 32 GB RAM.

Full instructions: [docs/Castelnuovo_Run_On_Desktop.pdf](#), which is inside the repository once it is cloned.

1. Carry on a USB stick — one file only

☐	File	Size	Where it is now
☐	<code>castelnuovo-image.tar.gz</code> the Docker image: OpenSees 3.8.0 + MUMPS, gms, the whole Python environment	1.40 GB 1 496 262 271 bytes	<code>C:\Users\mlaur\Desktop\</code>

Verify the copy before trusting it. A 1.4 GB USB transfer can truncate silently. On the desktop, in PowerShell, in the folder where you copied it:

```
Get-FileHash castelnuovo-image.tar.gz -Algorithm SHA256
```

The hash must be exactly:

```
FF893903453EBC636B3CD8C8B62D29FF8B5E44B2CE6F3F1286F8EEC52D468089
```

PowerShell prints it in uppercase; the comparison is case-insensitive. If it differs, copy the file again — do not proceed.

2. Nothing else needs carrying — it all comes from GitHub

☐	What	Included in the clone
☐	Analysis code, geometry, material data	yes
☐	Seismic record — Montenegro 1979, Run 2.1 longitudinal, PGA 0.593 g <code>resources/records/castelnuovo/montenegro1979_run21_long.txt</code> — already in m/s ² , do not rescale	yes
☐	Reference values for the verification step <code>resources/reference/castelnuovo_tied_10modes.json</code>	yes
☐	The step-by-step guide (PDF) <code>docs/Castelnuovo_Run_On_Desktop.pdf</code>	yes

```
git clone https://github.com/mlaura1996/modularBIMtoFEM.git
cd modularBIMtoFEM
git checkout chapter7-castelnuovo
```

The branch is `chapter7-castelnuovo`, not `main`. Last commit as pushed: `67dfac8`.

3. To install on the desktop (needs internet, ~15 min)

☐	Software	Notes
☐	Docker Desktop	Accept the WSL 2 backend; reboot; start it and wait for the whale icon to stop animating
☐	Git for Windows	All defaults
☐	File <code>C:\Users\<user>\.wslconfig</code>	Contents below. Then <code>wsl --shutdown</code> and restart Docker Desktop. Skipping this is the usual cause of the analysis being killed with no error message.

```
[wsl2]
memory=24GB
```

```
processors=8
swap=8GB
```

4. Free disk space needed on the desktop

Item	Space
Docker image once loaded (it decompresses)	~7 GB
Repository clone	~1 GB
Analysis output (recorders, sampled)	allow 20–30 GB
Total, with margin	~50 GB

5. Order of operations — do not reorder

1. Install Docker Desktop + Git, write `.wslconfig`, restart.
2. `docker load -i <path>\castelnuovo-image.tar.gz`
3. Clone the repository and check out `chapter7-castelnuovo`.
4. **Run the verification** — `verify_environment.py`, ~20 min. It must print `PASS`. Do not skip it.
5. **Run the smoke test** — `timehistory_castelnuovo.py --smoke`, ~10 min. Two lines matter; see below.
6. Only then, the full run, detached.

6. Two things to report back before the full run starts

(a) The tie check. Before the elements are built, the run prints `invariant check passed`. If it prints `ABORT: ... tie(s)` reference a node that the elements of their own volume do not use instead, stop and report it — it would be a bug in the script, not something to work around.

(b) The recorder check. Find the block `--- recorder check after 5 steps ---` and send back the list of any recorder marked `EMPTY`. The damage-output response names for this material could not be verified in advance; OpenSees writes a file holding only the time column instead of raising an error, so a 30-hour run could otherwise finish with no damage data at all.

Expected run time: roughly 15–40 hours for the 19.5 s window (3900 steps). The first progress line of the transient gives a real estimate for that machine. The setup stages before it — meshing, then 20–60 minutes building one damage material per element — are normal and not a hang.

The 8 cores will not speed this up: the solver is sequential. And **there is no restart** — if the run is interrupted it begins again from zero, so disable Windows automatic restarts and sleep first.

After copying the image to the desktop, delete `castelnuovo-image.tar.gz` from the laptop Desktop — that drive is at 97% capacity.